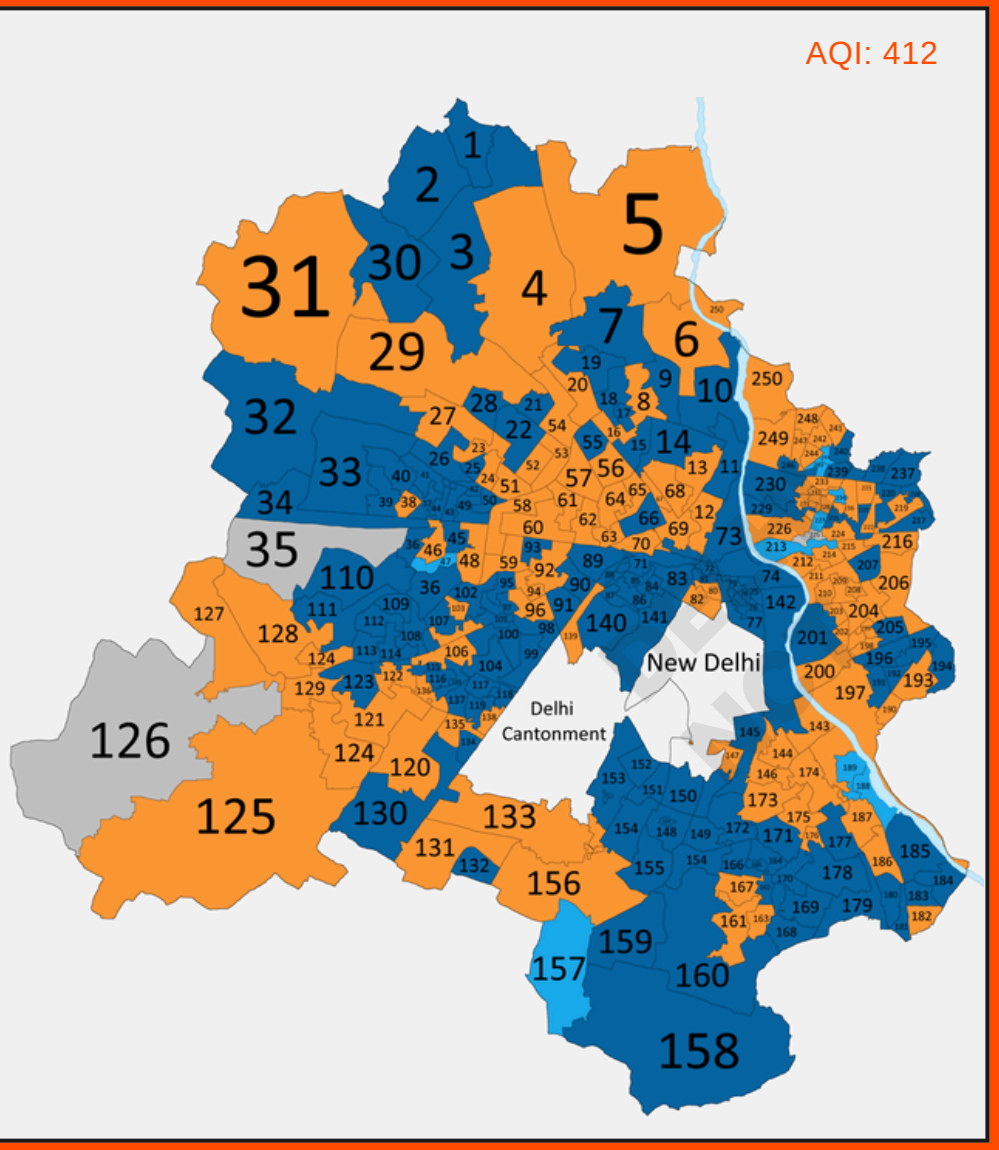
WARD-WISE INTELLIGENCE

A Hyper-Local Air Quality Intelligence System for Delhi NCT

"DECODING DELHI'S AIR, WARD BY WARD"



01 / DELHI'S AIR QUALITY CRISIS

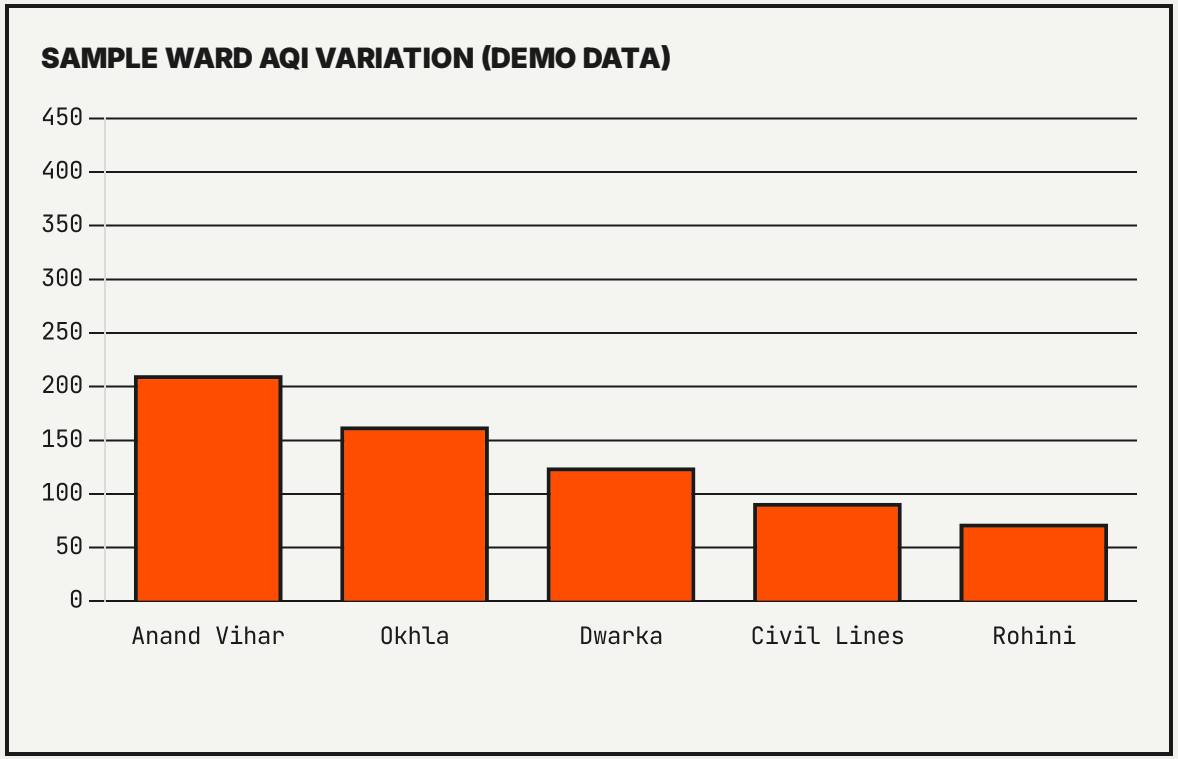
AQI 142-412

Delhi's 272 wards face varying pollution levels that current monitoring fails to capture with precision.

THE GRANULARITY GAP

Existing systems lack ward-level data, preventing targeted interventions where they are needed most.

Result: Inefficient resource allocation and generic health advisories.



02 / DUAL-INTELLIGENCE SYSTEM SOLUTION

CITIZEN INTELLIGENCE

HYPER-LOCAL MONITORING

Real-time AQI tracking across 1,240 sensor nodes for ward-specific precision.

ACTIONABLE HEALTH GUIDANCE

AI-driven personalized advisories based on WHO and CPCB standards.

CIVIC ACCOUNTABILITY

Incident reporting system with transparent status tracking and reward credits.

AUTHORITY INTELLIGENCE

COMMAND CENTER

Scientific ranking of wards by pollution severity and population exposure.

SOURCE ATTRIBUTION

AI-powered identification of pollution vectors (Industrial, Vehicular, etc.).

POLICY SIMULATION

Predictive impact assessment for interventions before city-wide implementation.

03 / EMPOWERING DELHI'S CITIZENS

HYPER-LOCAL AQI TRACKING

Access real-time pollution data specific to your ward. No more city-wide averages—get the data that affects your immediate neighborhood.

272 WARDS COVERED

AI HEALTH ADVISORY

Personalized protective actions based on current AQI levels. Integrated with WHO and CPCB guidelines to provide scientifically-backed safety advice.

REAL-TIME GUIDANCE

TRANSPARENT REPORTING

Report incidents with geotagged photos. Monitor the 4-stage pipeline from report to resolution with full transparency and accountability.

4-STAGE TRACKING

CITIZEN ENGAGEMENT: TRANSFORMING RESIDENTS INTO ACTIVE ENVIRONMENTAL PARTICIPANTS

04 / AUTHORITY COMMAND CENTER

PRIORITY QUEUE SYSTEM

Scientific ranking of wards by pollution severity, population exposure, and violation frequency.

WARD NAME	AQI	RISK LEVEL	STATUS
Anand Vihar	412	CRITICAL	Enforcing
Okhla Ph-III	320	SEVERE	Dispatched
Dwarka Sec-8	245	POOR	Monitoring
Civil Lines	180	MODERATE	Normal

MULTI-DEPARTMENTAL COORDINATION

TRAFFIC POLICE

NO-IDLING ENFORCEMENT

Targeted patrols in high-vehicular zones identified by AI.

MCD / DPCC

CONSTRUCTION DUST REGISTRY

Real-time monitoring of site compliance and smog gun deployment.

FIRE & SAN.

WASTE BURNING RESPONSE

Rapid response to citizen-reported biomass burning incidents.

SLA COMPLIANCE TRACKING

Automated alerts for mandate violations and response delays.

05 / AI POLICY SIMULATION ENGINE

POLICY INPUT

Test environmental interventions before city-wide rollout to optimize resource allocation.

SCENARIO A

"Deploy 12 Smog Guns in Anand Vihar"

SCENARIO B

"No-Entry for Heavy Transport"

SIMULATION PROCESS

AQI MODELING	Confidence 94%
FEASIBILITY	Logistical Score
EQUITY IMPACT	Vulnerability Check
ECONOMIC COST	ROI Projection

AI ANALYSIS

"No-Entry for heavy transport in Anand Vihar reduces localized PM10 by 18.4% in 6 hours."

PROJECTED IMPACT

-18.4%

LOCALIZED PM10 REDUCTION

SOCIAL EQUITY SCORE

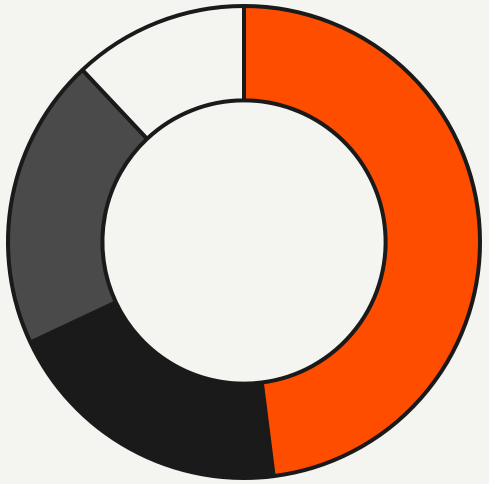
High protection for street vendors and children in high-density zones.

06 / EXPLAINABLE AI SOURCE ATTRIBUTION

POLLUTION FINGERPRINTING

Our AI models analyze chemical signatures and temporal patterns to identify the "Why" behind pollution spikes.

SOURCE DISTRIBUTION: ANAND VIHAR (DEMO)



Vehicular Industrial Biomass

AI EXPLAINABILITY ENGINE

Translating complex multi-variate data into plain-language insights for rapid decision-making.

CASE STUDY: ANAND VIHAR

AQI: 412 (CRITICAL)

"Peak congestion at ISBT and unauthorized biomass combustion in adjacent industrial zones are the primary drivers."

* Data synchronized with DPCC/CPCB reference stations.

07 / CLOSING THE ACCOUNTABILITY LOOP

4-STAGE TRACKING PIPELINE

Ensuring every citizen report is actioned with full transparency.

01 REPORTED (GEOTAGGED & PHOTO-VERIFIED)

02 ASSIGNED (AUTO-ROUTED TO AGENCY)

03 ACTIONED (FIELD ENFORCEMENT LOGGED)

04 RESOLVED (CITIZEN VERIFICATION)

CLEAN AIR CREDITS (CAC)

INCENTIVIZING CIVIC ACTION

A unique reward system for verified environmental contributions. Citizens earn credits for reporting violations and participating in community taskforces.

15 CAC = 1 kWh FREE EV CHARGING

SLA ENFORCEMENT

Automated deadline monitoring with escalation to senior officials for unresolved reports.

07 / TECHNICAL INFRASTRUCTURE

DATA LAYER

SENSOR NETWORK
1,240 IoT nodes across 272 wards with real-time telemetry.

INTEGRATION
Bi-directional sync with DPCC and CPCB reference stations.

STORAGE
Scalable MongoDB cluster for high-velocity environmental data.

AI ENGINE

CORE MODEL
Gemini AI for natural language generation and insight synthesis.

ATTRIBUTION
Custom neural networks for multi-variate source identification.

SIMULATION
Predictive modeling engine with 94% confidence intervals.

SYSTEM STACK

FRONTEND
React / TypeScript PWA. Mobile-first responsive design.

Vite | Tailwind | Chart.js

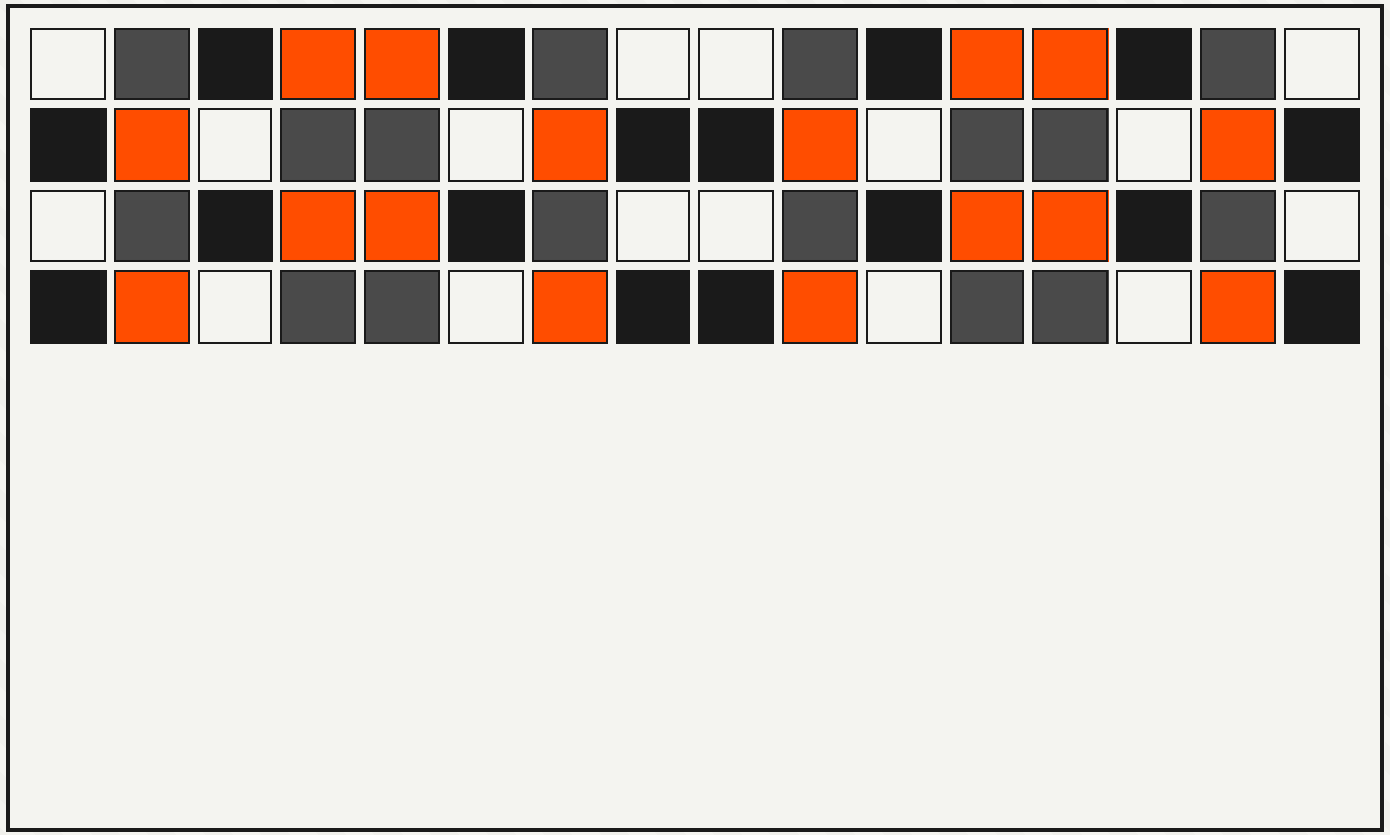
BACKEND
Node.js / Express. RESTful API with role-based security.

JWT | Redis | Docker

ARCHITECTURE: ROBUST, SCALABLE, AND SECURE CITY-WIDE DEPLOYMENT

09 / GEOGRAPHIC INTELLIGENCE HEATMAP

DELHI NCT: 272 WARD INTELLIGENCE GRID



CRITICAL SEVERE POOR MODERATE

INTERACTIVE OVERLAY

Visual representation of all 272 Delhi wards with real-time, color-coded AQI indicators.

CLICK-TO-EXPLORE

Deep dive into detailed ward profiles, including pollution fingerprints and trend visualization.

HOTSPOT TRACKING

Rapidly identify emerging pollution clusters and coordinate multi-departmental response.

11 / MEASURABLE IMPACT METRICS

2.4h

RESPONSE TIME

Average complaint resolution time, reduced from the previous 8-12 hour baseline.

85%

COMPLIANCE

Success rate in monitored construction sites through real-time dust registry tracking.

42

ACTIVE REPORTS

Daily verified incident reports in critical zones, driving rapid departmental response.

100%

TRANSPARENCY

Commitment to open-data for anonymized datasets, enabling public audit and research.

12 / ROADMAP AND FUTURE VISION

PHASE 01

PILOT LAUNCH

MONTHS 1-3

- Deploy in 4 high-priority wards (Anand Vihar, Okhla, Dwarka, Civil Lines).
- Sensor installation and baseline calibration.
- Authority training and onboarding.

PHASE 02

EXPANSION

MONTHS 4-6

- Expand to 50 wards across all 11 districts.
- Launch citizen awareness campaigns.
- Full integration with DPCC/CPCB systems.

PHASE 03

FULL SCALE

MONTHS 7-12

- Full 272-ward city-wide deployment.
- AI model fine-tuning with real-world data.
- Policy simulation validation with actual outcomes.

"A DELHI WHERE EVERY WARD BREATHES CLEAN AIR, EVERY CITIZEN HAS A VOICE, AND EVERY POLICY DECISION IS INFORMED BY SCIENCE."